

AN Mail Server

Linux Mail Server

Section 1: Service Brief

1.1 What is a Mail Server?

A **Mail Server** is a system responsible for **sending, receiving, storing, and providing access to email messages**.

A typical Linux mail server is built from different components, where each component performs a specific role.

Main Components

Component	Software	Main Role
MTA — Mail Transfer Agent	Postfix	Sends, receives, and transfers email
Mail Access Server	Dovecot	Allows users to access mail using IMAP/POP3
Webmail	Roundcube	Provides a browser-based interface for email
Mail Client	Thunderbird, Outlook, etc.	Used by users to send/read email

1.2 Postfix

Postfix is a **Mail Transfer Agent (MTA)** used to send and receive email messages.

Its main responsibilities are:

- Accepting outgoing email from users/applications
- Receiving incoming email
- Transferring email between mail servers
- Routing email to the appropriate destination
- Delivering mail to local mailboxes

Postfix mainly works with **SMTP**.

Important SMTP Ports

Port	Purpose
25	SMTP server-to-server mail transfer
587	SMTP message submission from mail clients
465	SMTP submission using implicit TLS

1.3 Dovecot

Dovecot provides users with access to their stored email through **IMAP** and **POP3**.

It is commonly used together with Postfix in Linux-based mail systems.

Its main responsibilities include:

- Providing **IMAP** access
- Providing **POP3** access when configured
- Authenticating users for mail access
- Accessing messages stored in the user's mailbox
- Working with Postfix for mail delivery/access integration

Important Ports

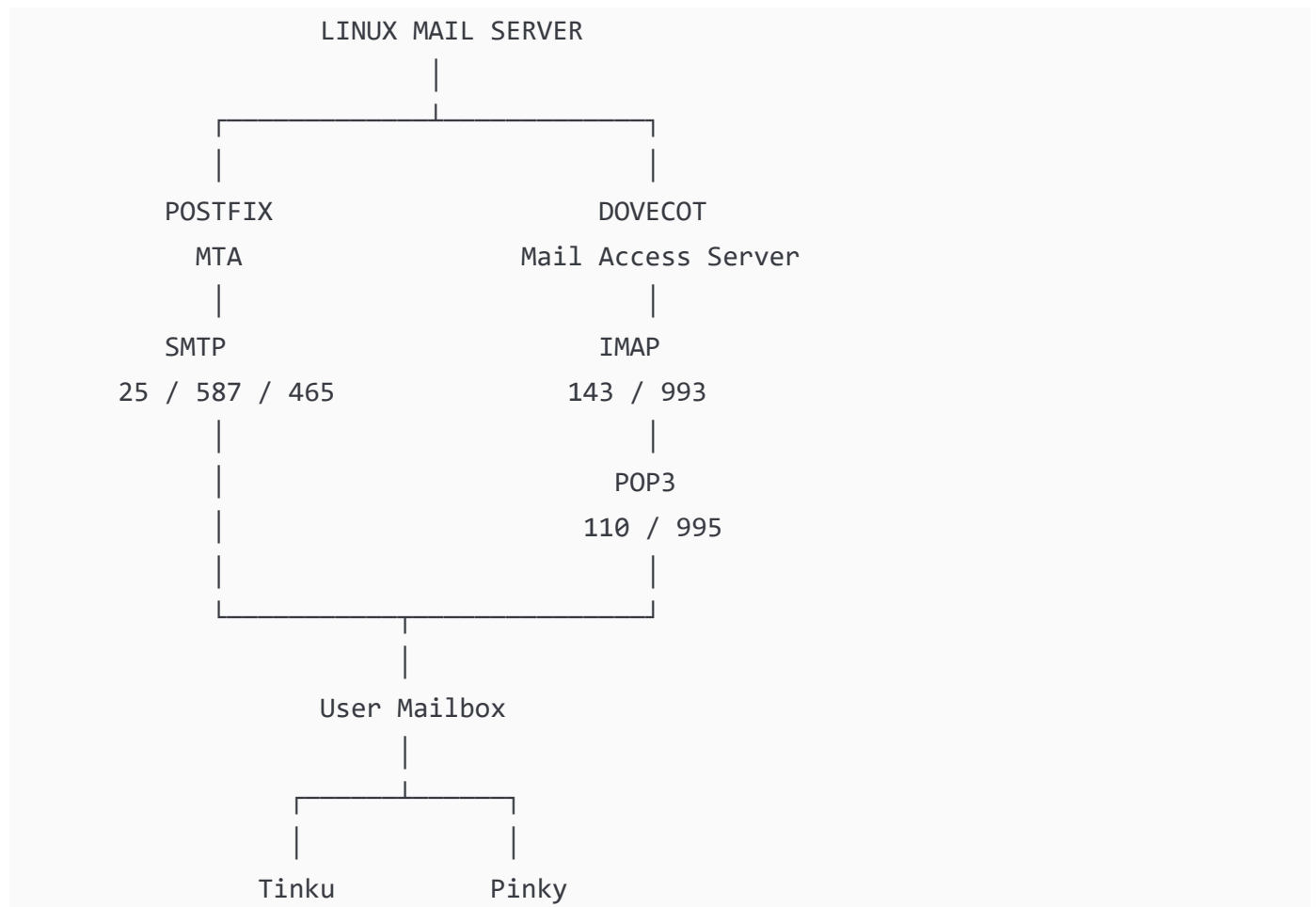
Protocol	Normal	TLS
IMAP	143	993
POP3	110	995

So, in simple terms:

```
Postfix → Handles SMTP / mail transfer
Dovecot → Handles IMAP/POP3 / user mail access
```

1.4 Postfix + Dovecot Together

Postfix and Dovecot perform different jobs, so they are commonly deployed together.



Postfix is responsible for moving the mail, while Dovecot provides the user's access to the mailbox.

1.5 What is Webmail?

A user does not necessarily need a command-line mail program to read email.

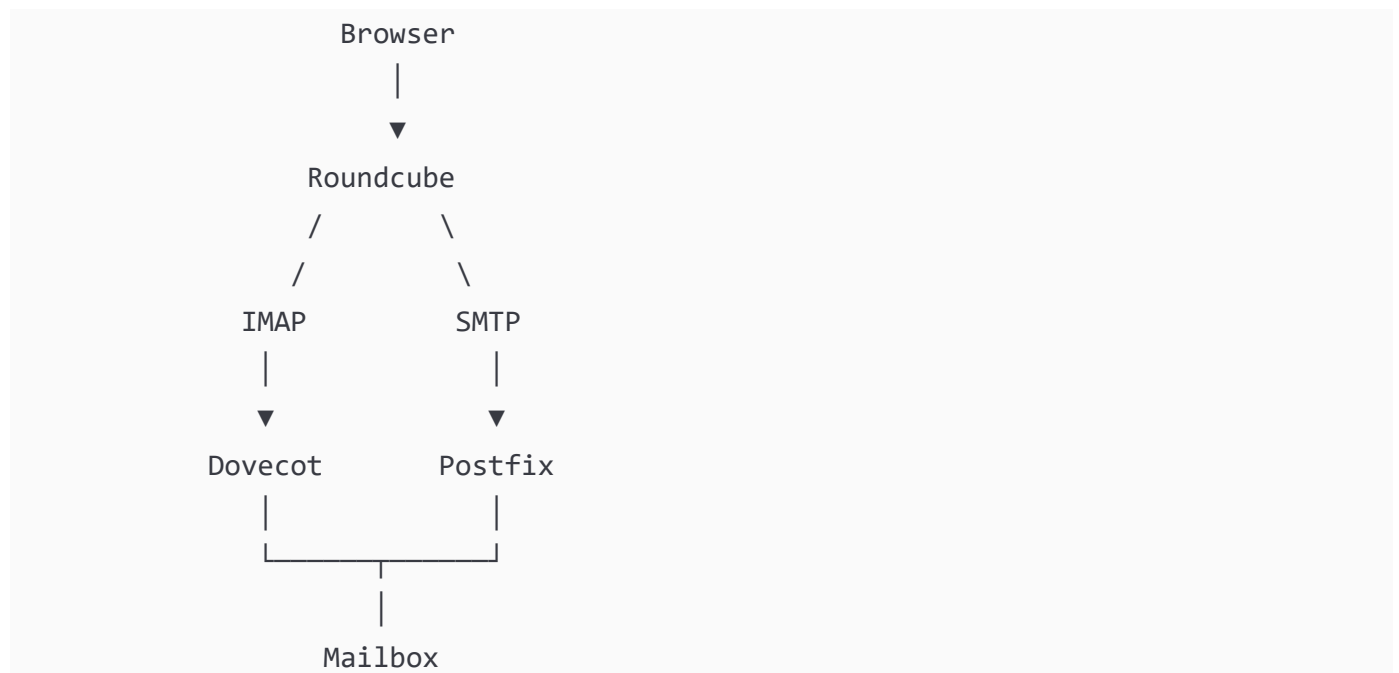
Webmail provides a browser-based interface for sending and reading messages.

Example: Roundcube

Roundcube is an open-source **webmail application**.

It is **not** the mail server itself.

Its role is:



So:

- **Postfix** → SMTP/mail transfer
 - **Dovecot** → IMAP/POP3/mail access
 - **Roundcube** → Web-based user interface
-

Section 2: Mail Protocols

The important protocols are:

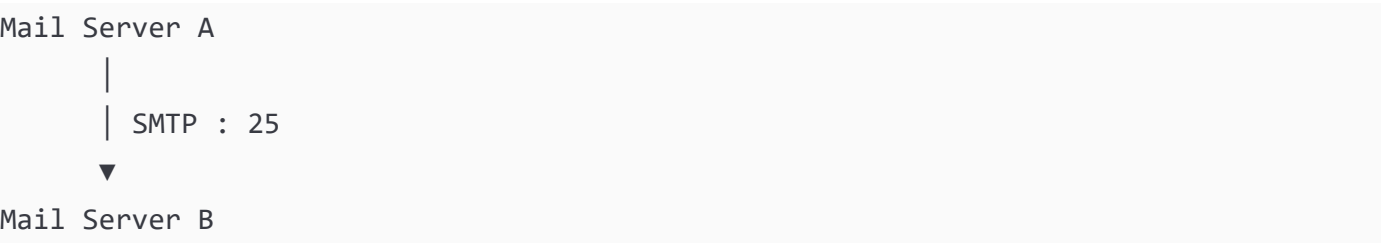
- **SMTP** — Sending and transferring email
- **IMAP** — Accessing and synchronizing email
- **POP3** — Retrieving email
- **TLS** — Securing mail communication

2.1 SMTP — Simple Mail Transfer Protocol

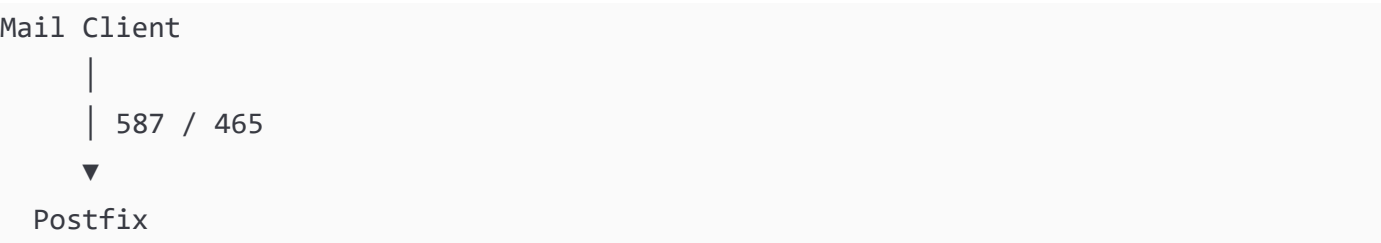
SMTP is used for **sending and transferring email**. Postfix primarily works as an **SMTP server (MTA)**.

Important SMTP Ports

Port	Purpose
25	Server-to-server SMTP mail transfer
587	Client mail submission
465	SMTP submission with implicit TLS



For a user's outgoing mail:



2.2 IMAP — Internet Message Access Protocol

IMAP is used by a mail client to **access and manage messages stored on the mail server**.

Dovecot provides the IMAP service.

IMAP Ports

Port	Purpose
143	IMAP
993	IMAP over TLS

IMAP is particularly useful when the same mailbox is accessed from multiple devices because messages and mailbox state are maintained on the server.

2.3 POP3 — Post Office Protocol Version 3

POP3 is used to **retrieve email from a mail server**.

Dovecot can provide POP3 when POP3 support is installed and configured.

POP3 Ports

Port	Purpose
110	POP3
995	POP3 over TLS

POP3 is traditionally designed around retrieving/downloading messages, whereas IMAP is designed for server-side mailbox access and synchronization.

2.4 IMAP vs POP3

Feature	IMAP	POP3
Main purpose	Access and synchronize mailbox	Retrieve/download mail
Normal port	143	110
TLS port	993	995
Mail remains managed on server	Yes	Traditionally downloaded
Multiple devices	Well suited	Less suited

Simple Memory

IMAP → Work with mail on the server

POP3 → Retrieve mail from the server

2.5 TLS — Transport Layer Security

TLS provides encryption for mail communication.

TLS is **not a separate mail protocol**. It is a security layer used with protocols such as SMTP, IMAP and POP3.

Examples:

SMTP + TLS

IMAP + TLS

POP3 + TLS

Important secure ports:

SMTP Submission → 465

IMAP → 993

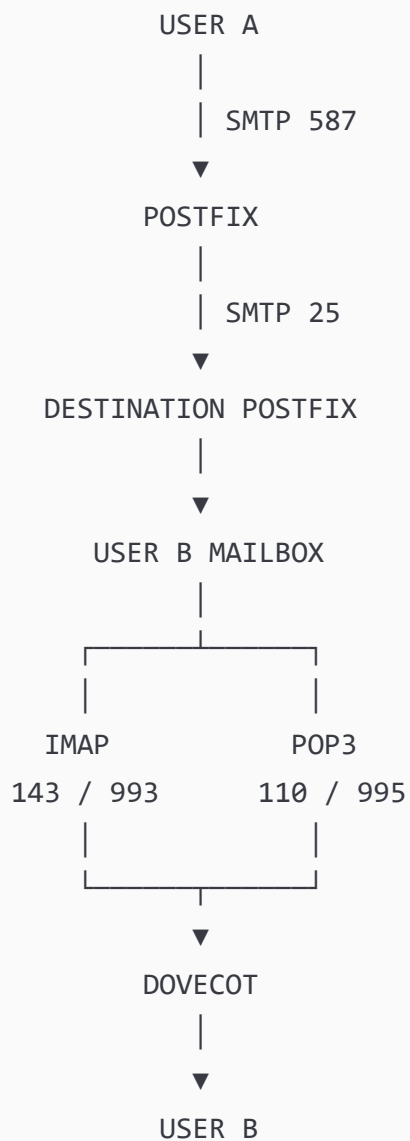
POP3 → 995

TLS helps protect:

- User credentials
 - Email contents
 - Communication between mail client and server
-

2.6 Complete Mail Protocol Flow

A simplified mail flow is:



SMTP → Send

IMAP → Access & Synchronize

POP3 → Retrieve

TLS → Secure

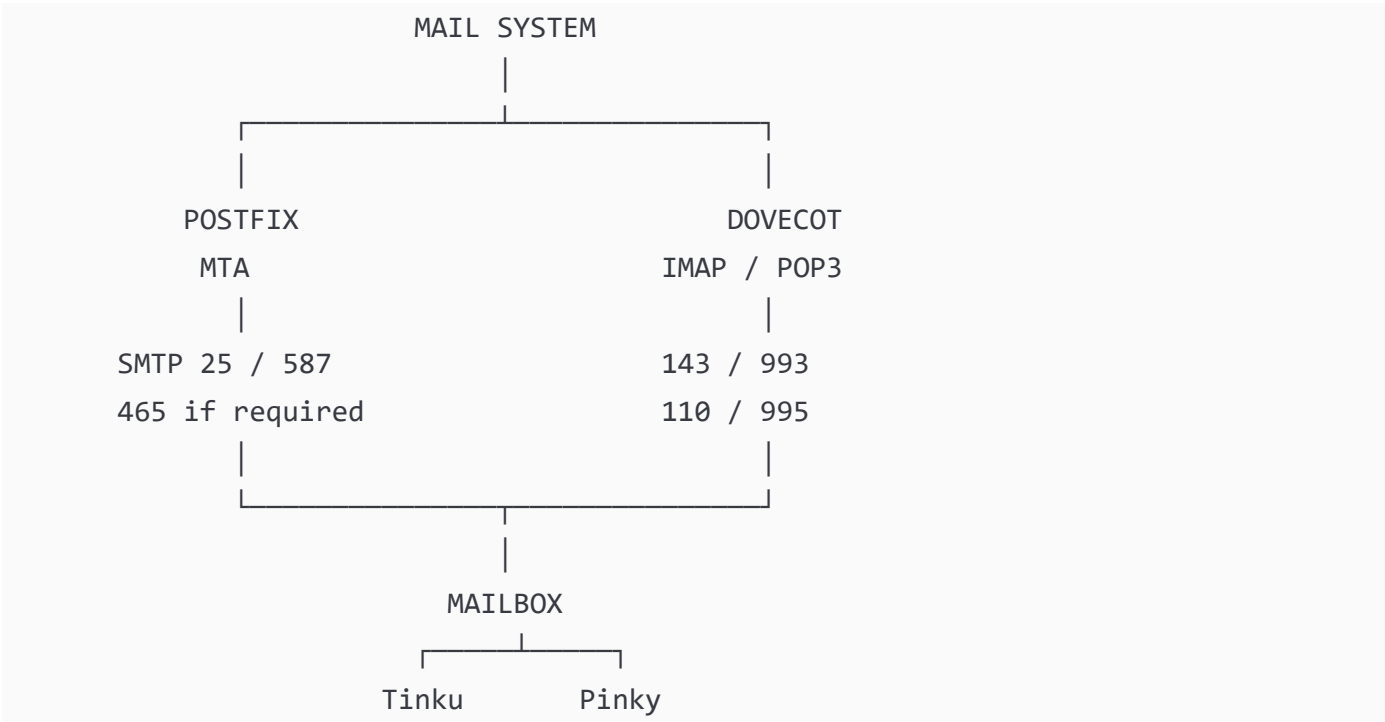
Section 3: Mail Server Architecture

3.1 Main Components

A Linux mail server consists of different components, each handling a specific part of the email process.

Component	Example	Main Responsibility
MUA / Mail Client	Thunderbird, Outlook	User sends and reads email
MTA	Postfix	Sends, receives and transfers email
Mail Access Server	Dovecot	Provides IMAP/POP3 access
Mailbox	User mail storage	Stores received messages
Webmail	Roundcube	Provides browser-based email access

3.2 Architecture



The main relationship is:

Postfix handles mail transfer, while Dovecot provides users access to their stored mail.

3.3 Postfix — MTA

Postfix acts as the **Mail Transfer Agent (MTA)**.

Its main responsibilities are:

- Accept outgoing email
- Receive incoming email
- Process and route messages
- Transfer email to another mail server
- Deliver mail to local mailboxes

Postfix primarily uses **SMTP**.

Important SMTP ports:

- **25** — server-to-server mail transfer
 - **587** — client mail submission
 - **465** — SMTP submission with implicit TLS
-

3.4 Dovecot — Mail Access Server

Dovecot provides users access to their stored email through **IMAP and POP3**.

Its main responsibilities are:

- Provide IMAP access
- Provide POP3 access when configured
- Authenticate mail users
- Allow users to access messages in their mailbox

Important ports:

- **143 / 993** — IMAP
- **110 / 995** — POP3

Dovecot therefore works mainly on the **mailbox access side**, while Postfix works on the **mail transfer side**.

3.5 Local Mail Flow

For example:

```
echo "Hello, Pinky" | mail -s "Test email subject" pinky
```

The basic process is:

1. **Tinku** sends the message.
2. **Postfix** processes the email.
3. The message is delivered to **Pinky's mailbox**.
4. **Dovecot** provides access to the mailbox through IMAP/POP3.
5. **Pinky** can read the message.

For basic local testing, the `mail` command can also be used to check whether local mail delivery is working.

Section 4: Postfix Installation and Basic Configuration

4.1 Install Postfix

Update the package repository:

```
sudo apt update
```

Install Postfix:

```
sudo apt install postfix -y
```

During installation, select:

```
Local only
```

4.2 Install Mailutils

Install the `mail` command for testing:

```
sudo apt install mailutils -y
```

4.3 Start and Verify Postfix

Check the service:

```
systemctl status postfix
```

If required, start and enable it:

```
sudo systemctl enable postfix
```

Verify that SMTP port **25** is listening:

```
ss -antp | grep :25
```

4.4 Create Local Mail Users

Create two users for testing:

```
sudo adduser tinku
```

```
sudo adduser pinky
```

4.5 Send a Local Test Email

Send an email to Pinky:

```
echo "Hello, Pinky" | mail -s "Test email subject" pinky
```

Part	Meaning
<code>echo</code>	Provides the message body
<code>mail</code>	Sends the email
<code>-s</code>	Specifies the subject
<code>pinky</code>	Recipient

4.6 Read the Mail

Switch to Pinky's account:

```
su pinky
```

Open the mailbox:

```
mail
```

Useful commands inside `mail`:

```
?      → Show available commands
1      → Open message 1
d 1    → Delete message 1
q      → Quit
```

4.7 Test the Reverse Direction

Send a reply/test message to Tinku:

```
echo "Hello tinku, Good night" | mail -s "Return mail" tinku
```

Switch to Tinku:

```
su tinku
```

Read the mailbox:

```
mail
```

This confirms that **local users can send and receive mail through Postfix.**

Section 5: Dovecot Installation and Basic Configuration

5.1 Install Dovecot

Install the Dovecot core package and IMAP support:

```
sudo apt install dovecot-core dovecot-imapd -y
```

If POP3 access is also required:

```
sudo apt install dovecot-pop3d -y
```

Packages

Package	Purpose
dovecot-core	Core Dovecot service and configuration
dovecot-imapd	IMAP support
dovecot-pop3d	POP3 support

5.2 Check Dovecot Service

Check whether Dovecot is running:

```
systemctl status dovecot
```

If required, start and enable it:

```
sudo systemctl enable dovecot
```

5.3 Verify IMAP and POP3 Ports

Check the listening ports:

```
ss -antp | grep -E ' :143|:993|:110|:995 '
```

Depending on which protocols are enabled, you should see the corresponding ports:

Protocol	Port
IMAP	143
IMAPS	993
POP3	110

Protocol	Port
POP3S	995

5.4 Dovecot Configuration Files

Dovecot uses its configuration files under:

```
/etc/dovecot/
```

Important configuration files include:

```
/etc/dovecot/dovecot.conf
/etc/dovecot/conf.d/
```

The `conf.d` directory contains separate configuration files for different Dovecot components.

For example:

```
/etc/dovecot/conf.d/
├─ 10-auth.conf
├─ 10-mail.conf
├─ 10-master.conf
└─ 10-ssl.conf
```

Their general purpose is:

File	Purpose
<code>10-auth.conf</code>	Authentication settings
<code>10-mail.conf</code>	Mailbox location/settings
<code>10-master.conf</code>	Dovecot service/listener settings
<code>10-ssl.conf</code>	TLS/SSL settings

5.5 Mailbox Location

Dovecot needs to know where the user's mail is stored.

A common mailbox format is **Maildir**, for example:

```
~/Maildir/
```

Postfix must deliver mail to the location that Dovecot is configured to read.

The exact mailbox location must match the way Postfix delivers the mail, If these locations do not match, Postfix may successfully deliver the message while Dovecot does not show it to the user.

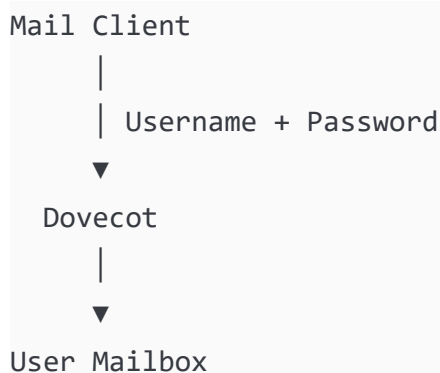
5.6 Authentication

Dovecot authenticates users when they connect through IMAP or POP3.

```
tinku
pinky
```

Their credentials are used when a mail client connects to Dovecot.

The basic relationship is:



5.7 Restart and Verify

After making Dovecot configuration changes:

```
sudo systemctl restart dovecot
```

Check the service:

```
systemctl status dovecot
```

Then verify the listening ports again:

```
ss -antp | grep -E ' :143|:993|:110|:995 '
```

Section 6: Mail Server Practical Testing

6.1 Create Test Users

If the users have not already been created:

```
sudo adduser tinku
sudo adduser pinky
```

These users will be used for testing local email delivery.

6.2 Send Mail to Pinky

Send a test email:

```
echo "Hello, Pinky" | mail -s "Test email subject" pinky
```

Here:

Part	Purpose
<code>echo</code>	Provides the email body
<code>mail</code>	Sends the email
<code>-s</code>	Specifies the subject
<code>pinky</code>	Recipient

6.3 Read Pinky's Mail

Switch to Pinky's account:

```
su pinky
```

Open the mailbox:

```
mail
```

The test message should be displayed.

Useful commands:

```
?      → Show available commands
1      → Read message 1
d 1    → Delete message 1
q      → Quit
```

6.4 Send Mail Back to Tinku

Now test the reverse direction:

```
echo "Hello tinku, Good night" | mail -s "Return mail" tinku
```

Switch to Tinku:

```
su tinku
```

Read the mailbox:

```
mail
```

The returned message should be displayed.

6.5 Practical Result

The practical is successful when:

- Tinku can send mail to Pinky.
- Pinky can read the received mail.
- Pinky can send mail back to Tinku.
- Tinku can read the returned mail.

This confirms that **local email delivery through Postfix is working**.

Section 7: Troubleshooting

Troubleshooting should focus on common practical mistakes and the checks used to locate the problem.

7.1 Service Not Running

Check both services:

```
systemctl status postfix
systemctl status dovecot
```

If required:

```
sudo systemctl restart postfix
sudo systemctl restart dovecot
```

Check logs:

```
sudo journalctl -u postfix
sudo journalctl -u dovecot
```

7.2 Required Port Not Listening

Check SMTP:

```
ss -antp | grep :25
```

Check IMAP/POP3:

```
ss -antp | grep -E ' :143|:993|:110|:995 '
```

If a required port is missing:

- Check the corresponding service.
- Check its configuration.
- Restart the service.
- Check the logs.

7.3 Mail Sent but Not Received

If the test mail is not received:

```
echo "Hello, Pinky" | mail -s "Test email subject" pinky
```

Check:

1. Postfix status

```
systemctl status postfix
```

2. Postfix logs

```
sudo journalctl -u postfix
```

3. Correct recipient mailbox

```
su pinky  
mail
```

4. Mailbox location

Postfix must deliver mail to a location that Dovecot is configured to access.

7.4 Mailbox Empty in Dovecot

If Postfix delivers the message but Dovecot does not show it, check the mailbox configuration:

```
sudo nano /etc/dovecot/conf.d/10-mail.conf
```

The common problem is:

```
Postfix → Mailbox A
```

```
Dovecot → Looking at Mailbox B
```

Both must use the correct mailbox location.

7.5 Quick Troubleshooting Order

```
Service status
```

```
↓
```

```
Listening ports
```

```
↓
```

```
Configuration
```

```
↓
```

```
Mailbox location
```

```
↓
```

```
Logs
```

```
↓
```

```
Test mail again
```

Section 8: Quick Reference

8.1 Important Packages

Package	Purpose
<code>postfix</code>	SMTP mail server / MTA
<code>dovecot-core</code>	Dovecot core service
<code>dovecot-imapd</code>	IMAP support
<code>dovecot-pop3d</code>	POP3 support
<code>mailutils</code>	Command-line mail testing

8.2 Important Ports

Service	Port	Purpose
SMTP	25	Server-to-server mail transfer
SMTP Submission	587	Client mail submission
SMTP TLS	465	Secure SMTP submission
IMAP	143	Mailbox access
IMAPS	993	IMAP over TLS
POP3	110	Mail retrieval
POP3S	995	POP3 over TLS

Easy Memory

25 → SMTP / Mail Transfer
587/465 → SMTP / Submission
143/993 → IMAP
110/995 → POP3

8.3 Important Configuration Locations

Postfix

`/etc/postfix/`

Main configuration:

`/etc/postfix/main.cf`

Dovecot

```
/etc/dovecot/
```

Important files:

```
/etc/dovecot/dovecot.conf
/etc/dovecot/conf.d/10-auth.conf
/etc/dovecot/conf.d/10-mail.conf
/etc/dovecot/conf.d/10-master.conf
/etc/dovecot/conf.d/10-ssl.conf
```

8.4 Important Commands

Service Management

```
systemctl status postfix
systemctl status dovecot

sudo systemctl restart postfix
sudo systemctl restart dovecot
```

Check Ports

```
ss -antp
```

Check Logs

```
sudo journalctl -u postfix
sudo journalctl -u dovecot
```

Local Mail Testing

```
echo "Hello, Pinky" | mail -s "Test email subject" pinky

su pinky
mail
```

8.5 Postfix and Dovecot — Quick Difference

Postfix	Dovecot
MTA	Mail Access Server
SMTP	IMAP / POP3
Sends/transfers mail	Provides mailbox access
Main port 25	IMAP 143/993 , POP3 110/995

Section 9: Viva / Interview Questions

1. What is Postfix?

Postfix is an **MTA** used for sending, receiving, routing and delivering email.

2. What is Dovecot?

Dovecot provides **IMAP/POP3 access** to users' mailboxes.

3. What is SMTP?

Protocol used for sending and transferring email.

4. What is SMTP port 25?

It is mainly used for **server-to-server mail transfer**.

5. What are IMAP ports?

143 for IMAP and **993** for IMAP over TLS.

6. What are POP3 ports?

110 for POP3 and **995** for POP3 over TLS.

7. How do you check Postfix and Dovecot?

```
systemctl status postfix
systemctl status dovecot
```

8. How do you check listening ports?

```
ss -antp
```

9. How do you send a test email?

```
echo "Hello, Pinky" | mail -s "Test email subject" pinky
```

10. How do you read the mailbox?

```
su pinky
mail
```

11. What if mail is delivered but not visible through Dovecot?

Check the **mailbox location** configured for Postfix and Dovecot.

12. What is the main difference between Postfix and Dovecot?

Postfix handles mail transfer; Dovecot provides mailbox access.